MA341
Applied ODE's

Final Review

1. Solve the following homogeneous problem $x'(t) = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} x(t)$

Solution:

$$x(t) = c_1 e^t \begin{pmatrix} \cos 2t \\ \cos 2t + \sin 2t \end{pmatrix} + c_2 e^t \begin{pmatrix} \sin 2t \\ \sin 2t - \cos 2t \end{pmatrix}$$

2. Solve the following systems of ODEs subject to given initial conditions.

a. $x'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t) + \begin{pmatrix} e^t \\ t \end{pmatrix}$, $x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Solution:

$$x(t) = \frac{1}{4} \begin{pmatrix} 5 \\ 15 \end{pmatrix} e^{-t} + \frac{1}{4} \begin{pmatrix} -1 + 6t \\ -3 + 6t \end{pmatrix} e^t + \begin{pmatrix} t \\ 2t - 1 \end{pmatrix}$$

b. $x'(t) = \begin{pmatrix} 5 & 3 \\ -6 & -4 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ e^t \end{pmatrix}$, $x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Solution:

$$x(t) = \frac{1}{2} \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} + \begin{pmatrix} 4 \\ -4 \end{pmatrix} e^{2t} + \frac{1}{2} \begin{pmatrix} -3 \\ 4 \end{pmatrix} e^t + \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

c. $x'(t) = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} x(t) + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$, $x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Solution:

$$x(t) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} (t-1) \sin t + \begin{pmatrix} t \\ t + 1 \end{pmatrix} \cos t$$

3. Solve the following problem

$$\begin{cases} y\cos(x+y)dx + (3\sin(x+y) + y\cos(x+y))dy = 0 \\ y(0) = \pi/2 \end{cases}$$

Solution: This not an exact equation, but a integration factor can be found. The solution is

$$y^3 \sin(x+y) = \frac{\pi^3}{8}.$$

4. Consider

$$xy'' - (1+x)y' + y = 0$$

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a. Given $y_1 = e^x$, find the second linearly independent solution.

Solution:

$$y_2 = -(x+1)$$

b. Use variation of parameters to solve $xy'' - (1+x)y' + y = x^2e^{2x}$

Solution:

$$y = c_1e^x + c_2(x+1) + \frac{1}{2}e^{2x}(x-1)$$

5. Solve the following initial value problem

$$\begin{cases} y'' - 5y' + 6y = 2\delta(t-1) + 3\delta(t-4) \\ y(0) = 0 \\ y'(0) = 0 \end{cases}$$

Solution:

$$y(t) = 2u_1(t)(e^{3(t-1)} - e^{2(t-1)}) + 3u_4(t)(e^{3(t-4)} - e^{2(t-4)})$$

6. Solve the following initial value problem

$$\begin{cases} dy + (\frac{2y}{x} - \frac{y^3}{x^2})dx = 0 \\ y(1) = 1 \end{cases}$$

Solution: Rewrite the equation as Bernuli's $x^2y' + 2xy = y^3$.

$$y = 0 \quad \text{or} \quad y = \left(\frac{2}{5x} + Cx^4\right)^{-1/2}$$

Use IC to get $C = \frac{3}{4}$.

7. Consider

$$y'' - 2y' + y = (x+4)e^x$$

a. Find a general solution.

Solution:

$$y_h = c_1e^x + c_2xe^x$$

$$y_p = \left(\frac{1}{6}x^3 + 2x^2\right)e^x$$

$$y = y_h + y_p == c_1e^x + c_2xe^x + \left(\frac{1}{6}x^3 + 2x^2\right)e^x$$

b. Find a solution satisfying the ICs $y(0) = 1, y'(0) = 1$.

Solution: $C_1 = 1, C_2 = 0$, so

$$y = e^x + \left(\frac{1}{6}x^3 + 2x^2\right)e^x$$